Uncertainty-Aware Bayesian Mineral Prospectivity Mapping under Spatial Separation: Heterogeneous Predictor Relationships across the Central African Copperbelt

Abstract

Whether geological and geophysical predictor relationships remain predictive across geographically separated portions of a mineralized belt, and how those relationships vary spatially, is a central question for regional mineral prospectivity mapping. We address this question in the Central African Copperbelt using a Domain-Varying Hierarchical Model (DVHM), a partially pooled Bayesian logistic regression evaluated across four geographically contiguous held-out sectors. A compact global logistic baseline (CGLB) provides a comparator. The 1,872-cell modelling frame (nominal 5 km × 5 km grid; 138 cells with a compiled occurrence record) includes distance to mapped faults, distance to lithological contacts, Bouguer gravity, and retained surficial-lithology classes as predictors. Intercepts and the four distance coefficients (linear and quadratic, constructed from standardized distances) vary by Daly tectonic domain, while Bouguer gravity and surficial-lithology effects are global. Posterior draws provide uncertainty distributions for cell-level conditional occurrence probabilities. Predictor contributions to the out-of-fold linear predictor vary markedly across along-belt sectors: fault-distance contribution is highest in the northern sectors, while Bouguer gravity becomes increasingly important toward the south. Across four distance-representation specifications, mean ROC-AUC ranges from 0.671 to 0.686 and mean PR-AUC from 0.107 to 0.145. Distance associations recur across representations, whereas quadratic curvature is representation-dependent and the population-level stationary-point diagnostic D^* is not identifiable. The analysis therefore reveals spatially heterogeneous fitted relationships between geological and geophysical predictors and prospectivity, together with heterogeneous geographic predictive transferability across the Copperbelt. These patterns are consistent with a spatially heterogeneous geological framework, while the statistical associations alone do not establish a specific causal geological mechanism or a universal distance relationship.

Keywords: mineral prospectivity mapping; Bayesian hierarchical model; spatial cross-validation; spatial heterogeneity; Central African Copperbelt; Daly tectonic domains; parameter uncertainty

1 Introduction

Regional mineral prospectivity mapping integrates geological, geochemical, and geophysical evidence to identify locations associated with known deposits. Published approaches range from knowledge-driven evidence integration (Bonham-Carter, 1994; Carranza, 2009) to logistic regression, weights of evidence (Bonham-Carter et al., 1989), and machine-learning classifiers (Porwal et al., 2006; Carranza and Laborte, 2015; Ford, 2019; Yousefi et al., 2021; Matthieu et al., 2026). A fundamental question is whether predictor–prospectivity relationships estimated in one part of a mineralized belt remain predictive in geographically separated sectors—and if not, how they vary

spatially. In the Central African Copperbelt, regional variation in basin structure, stratigraphy, and ore assemblages (Cailteux et al., 2005; Selley et al., 2005; Hitzman et al., 2012; Key et al., 2004; Ndonfack et al., 2025) suggests that a single set of global predictor weights may not adequately capture along-belt heterogeneity. Standard random cross-validation can underestimate prediction error when spatially proximate observations share covariates and occurrence histories (Brenning, 2012; Roberts et al., 2017); the magnitude of this bias depends on the prediction target and blocking design, and spatial blocks can create predictor extrapolation (Valavi et al., 2019).

We address this question using the Domain-Varying Hierarchical Model (DVHM), a partially pooled Bayesian logistic regression in which intercepts and distance-predictor coefficients vary by Daly tectonic domain (Daly et al., 2025), while Bouguer gravity and surficial-lithology effects are shared globally. Spatially varying coefficient structures provide a statistical representation of changing relationships without asserting causation (Gelfand et al., 2003; Fotheringham et al., 2002). The Bayesian formulation represents posterior uncertainty in parameters, hierarchical scales, and domain effects; for each held-out cell, posterior draws produce a distribution of conditional occurrence probabilities $p_i^{(s)} = \text{logit}^{-1}(\eta_i^{(s)})$, summarized by the posterior mean, median, standard deviation, and 95% credible interval. Uncertainty in evidence layers, model parameters, conditional probabilities, and outcomes are distinct quantities in prospectivity work (Lisitsin et al., 2014; Parsa and Carranza, 2021; Zuo et al., 2021). Transfer is evaluated across four geographically contiguous along-belt sectors rather than through random cell splitting.

This study contributes (i) a partially pooled hierarchical Bayesian prospectivity framework evaluated under along-belt spatial separation, and (ii) a representation-sensitivity design that separates recurring proximity associations from representation-dependent polynomial curvature and D^* stationary-point diagnostics. We ask how fault distance, lithological-contact distance, Bouguer gravity, and surficial lithology jointly relate to recorded occurrence labels; how model-component contributions differ across held-out sectors; how out-of-fold discrimination varies; and whether distance-response conclusions persist under alternative mathematical representations.

2 Study area, spatial data, and evidence inventory

2.1 Study area and grid definition

The study region encompasses the Central African Copperbelt across the border region of northern Zambia and the southern Democratic Republic of the Congo. The spatial frame is a nominal $5\,\text{km} \times 5\,\text{km}$ grid in WGS 84 / UTM Zone 35S (EPSG:32735). A full 8,536-cell grid was generated and intersected with the union of the five USGS permissive-tract polygons, which are distributed in Arc 1950 Lambert Conformal Conic and were reprojected to EPSG:32735. The intersection retained 1,899 cells and clipped edge cells to the tract union; 253 cells retain the full $25\,\text{km}^2$ area. The retained cells cover $21{,}790\,\text{km}^2$ and span approximately 500 km along strike, with cell centroids from Easting 282,675 to 715,465 m and Northing 8,380,251 to 8,859,512 m. The permissive tracts define the assessment frame; they are not a geological contact or the limit of mineralization.

The regional geological context and compiled occurrence locations are shown in Figure 1.

The observational unit is one nominal $5\,\text{km} \times 5\,\text{km}$ grid cell, represented by its clipped polygon, centroid coordinates (x_i, y_i), and a binary compiled-record indicator $Y_i \in \{0, 1\}$. The prepared table contains 1,899 cells and 139 positive labels. Twenty-seven cells have no stored domain assignment, including one positive cell; complete-case filtering and removal of unrecognized domain

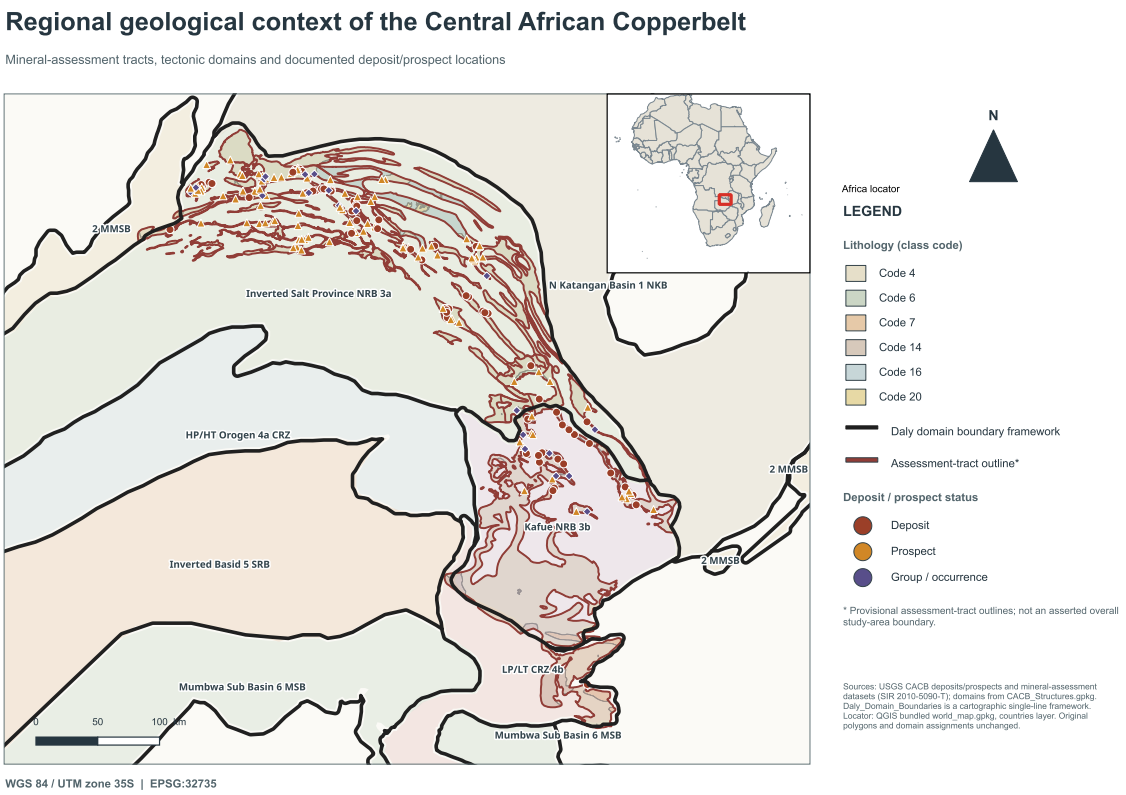


Figure 1: Regional geological context for the Copperbelt modelling frame. The map shows the Daly tectonic domains, the USGS permissive-tract outlines used to define the modelling frame, mapped surficial lithology, and compiled deposit and prospect locations. Occurrence and tract data are attributed to the U.S. Geological Survey (USGS) Central African Copperbelt assessment (Zientek et al., 2014); tectonic-domain context follows Daly et al. (2025), and the mapped structural compilation is attributed to Selley et al. (2005). The lithology background is the Africa surficial lithology compilation (Sayre and Kabuchanga, 2015); its metadata describe a mix of bedrock and unconsolidated parent materials and note that spatial complexity and classification detail vary across Africa.

labels therefore yield 1,872 cells, including 138 positive and 1,734 background cells.

The response is cell-level presence, not an occurrence count: a cell is positive when at least one compiled USGS occurrence record falls within its polygon, and repeated records within a cell do not increase its label. The USGS Central African Copperbelt assessment database contains 187 point records: 102 prospects, 65 deposits, and 20 grouped records; grouped records represent multiple sites within 500 m aggregated for resource modeling. Point-in-polygon assignment places 184 records inside the 1,899-cell frame and three outside; the 184 records map to 139 unique positive cells, of which 107 contain one record and the remainder contain two to five. Background cells indicate that no compiled record falls in the cell, not a verified absence of mineralization. The source metadata report a horizontal positional accuracy of 1,000 m, source-map scales from 1:5,000 to 1:2,000,000, manual digitization for some sites, and a use constraint that the database not be displayed at scales larger than 1:1,000,000. A 1,000 m positional uncertainty is one-fifth of the 5 km cell size, so records near cell edges can be assigned to adjacent cells under a different location interpretation. Accordingly, fitted probabilities describe the conditional probability of a compiled cell label under the model and validation design, not a probability of discovery.

2.2 Data sources and feature extraction

The predictors were assembled in QGIS and exported to the machine-learning framework. Table 1 summarizes the source-to-feature chain. The text below distinguishes steps that are confirmed by original source documentation or archived QGIS outputs from steps that are inferred from geometry, attributes, and project configuration.

Distance to mapped faults. The fault predictor is the Euclidean distance from each cell centroid to the nearest digitized fault line in EPSG:32735. The archived layer records one nearest fault per cell and attributes all lines to the Selley et al. (2005) Figure 1 compilation; the frozen values reproduce it to within 4.95×10^{-10} m. The fault raster carries a nine-point georeferencing fit recorded in a GDAL control-point file (fit residuals below 2 m in map coordinates), but the provenance of those control points and the positional accuracy of the underlying Figure 1 compilation are not stated in the available metadata. The available 126-line layer reproduces only 1,672 of the 1,899 frozen values, with the remaining 227 differing by up to about 33 km, so the exact fault geometry used for the frozen predictor is not preserved. The distance is a cartographic proximity measure, not a surveyed distance.

Surficial lithology and contact distance. The lithology source is the Africa surficial lithology raster (RCMRD SERVIR-Africa Team and USGS; 3 arc-second, WGS 84; values 0–20, comprising 19 lithology classes plus water and an n/a code; NoData 255). Its metadata describe a compilation and reclassification of twelve digital geology, soil, and lithology databases, mixing bedrock and unconsolidated parent materials; it is not a bedrock host-rock map, and its spatial complexity and classification detail vary across Africa. In QGIS the raster was reprojected to EPSG:32735 at 93 m and clipped to the tract union. The six classes present (metasedimentary, silicic, metaigneous, alluvium-fluvial, alluvium-saline, and water) match the class proportions of the source within the tract mask, consistent with nearest-neighbour resampling and no class reclassification; the exact warp options were not recorded. Polygonization produced 314 polygons; after conversion to lines and removal of the outer boundary, 259 internal contact lines remained.

Table 1: Source-to-feature provenance for the prepared 1,899-cell frame and the 1,872-cell complete-case modelling frame. "Confirmed" denotes numerical or documentary reproduction from an original source or archived QGIS output; "inferred" denotes an operation reconstructed from geometry, attributes, or project configuration but not recorded in the available processing history.

Feature (ML column)	Source product	Native support	QGIS-to-ML transformation and status
Distance to fault (`distance_to_fault`)	Fig. 1 structural compilation (Selley et al., 2005)	126 digitized fault lines; UTM 35S	Centroid-to-nearest-fault Euclidean distance. Frozen values reproduce the archived 1,899-cell distance layer exactly (maximum difference 4.95×10^{-10} m); the layer records one nearest fault per cell and the source field `Selley_et_al_2005_Fig1`. The available 126-line layer reproduces 1,672/1,899 frozen values; the other 227 differ by up to about 33 km, so the exact fault-geometry version is not preserved. Nine-point georeferencing fit recorded; source positional accuracy not stated in available metadata. Confirmed numerically; geometry version inferred.
Contact distance and class (`distance_to_lithology_contact`, `litho_contact_litho_class`)	Africa surficial lithology, RCMRD SERVIR-Africa/USGS (Sayre and Kabuchanga, 2015)	3 arc-second (∼93 m) categorical raster; WGS 84; values 0–20 (19 lithology classes plus water and n/a); NoData 255	Raster reprojected to EPSG:32735 at 93 m and clipped to the tract union; six classes present; 314 polygons; 259 internal contact lines. A many-to-many cell–contact join produced the archived candidate table. The frozen values are the first record per cell in that table (class exact; distance within 5×10^{-6} m), as implemented by the ML preparation's first-row retention. The first record is the minimum-distance candidate for only 636/1,899 cells; 1,263 differ (median +1.87 km, maximum +14.1 km). The archived class is attached to the contact feature rather than sampled at the cell centroid. Confirmed archive reproduction; selection order and contact-class semantics not recorded.
Bouguer gravity (`bouguer`)	BGI WGM2012 complete Bouguer anomaly (`WGM2012_Bouguer_ponc_2min`) (Bureau Gravimétrique International, 2012)	2 arc-minute grid; mGal; geographic lon/lat	Nearest-node sample at each cell centroid. All 1,899 frozen values match the source grid exactly; nearest-node offset is 69–2,549 m (median 1,471 m). No offset applied. Confirmed.
Tectonic domain (`domain`)	Tectonic framework (Daly et al., 2025); georeferenced Fig. 14 raster	11 digitized polygons and 45 boundary lines; UTM 35S	Centroid-in-polygon assignment. Stored labels match the archived centroid–domain table for all 1,899 cells, including 27 nulls. Against the original polygons, 69 centroids fall outside all polygons, two fall in overlaps, and five disagree; 49 complete-case cells are affected, all background. A diagnostic repaired layer leaves 27 null cells unmatched, zero complete-case unmatched, the same two overlaps, and no label disagreements. Confirmed label table; assignment and diagnostic repair inferred.
Occurrence response (`deposit_present`)	USGS SIR 2010-5090-T occurrence database (Zientek et al., 2014)	187 points; 102 prospects, 65 deposits, 20 groups; 1,000 m positional accuracy	Point-in-polygon to the 1,899 clipped cells. 184 records inside, three outside; 139 unique positive cells. Exact match to the frozen indicator. Background is no compiled record, not verified absence. Confirmed.
Grid and centroids (`id`, `centroid_x`, `centroid_y`)	5 km grid; USGS permissive-tract frame (Zientek et al., 2014)	8,536 full cells; EPSG:32735	Grid intersected with the five permissive tracts; 1,899 cells retained and edge cells clipped; 253 full 25 km^2 cells; total area 21,790 km^2. Confirmed.

The frozen class and distance reproduce the first record per cell in the archived `grid_features_litho_contact_v02.csv` table (class exact; distance within 5×10^{-6} m), because the ML preparation retained the first record per cell. This is not a nearest-contact rule: the first record is the minimum-distance candidate for 636 of 1,899 cells; 1,263 cells have a closer candidate (median +1.87 km, maximum +14.1 km), and 668 cells have a different class from the nearest candidate. Among 139 positive cells, 97 are affected by distance and 39 by class. The archived class is attached to the contact feature, not necessarily the lithology at the cell centroid. The frozen predictors were retained unchanged.

Bouguer gravity. The gravity predictor is the BGI WGM2012 complete Bouguer anomaly grid (`WGM2012_Bouguer_ponc_2min`; 2 arc-minute; mGal; geographic longitude/latitude). The QGIS project represents the grid as a delimited-text layer in EPSG:4326; each cell centroid was transformed from EPSG:32735 and assigned the nearest node. All 1,899 frozen values match the source exactly, with nearest-node offsets of 69–2,549 m (median 1,471 m). No offset was applied. The text header does not state a datum; EPSG:4326 follows the project configuration.

Tectonic domains. The six domain labels follow the Daly et al. (2025) tectonic framework; a georeferenced Figure 14 raster, an 11-polygon layer, and a 45-line boundary compilation are preserved. The stored labels match the archived centroid–domain table for all 1,899 cells, including 27 nulls; the frozen model uses the stored labels for the 1,872 complete-case cells. Against the original polygons, 69 centroids fall outside all polygons, two fall in overlaps, and five disagree with a single containing polygon. The 49 affected complete-case cells are all background, representing 2.8% of the 1,734 background cells; none of the 138 complete-case positive cells is affected. The only affected positive cell (id 7074) has a null domain and is outside the 1,872-cell complete-case frame. A repaired layer leaves the 27 nulls unmatched, has zero complete-case unmatched cells, retains the same two overlaps, and removes the five disagreements; it reproduces the stored labels for the 49 complete-case cells. These discrepancies are therefore best characterized as a cartographic linework limitation affecting some complete-case background cells without changing their stored labels; no complete-case positive assignment is affected. The repaired layer is a diagnostic reconstruction, not evidence of the original assignment, which is inferred to have been a centroid-in-polygon join with gap and overlap resolution.

Auxiliary layers. The archive also contains BGI WGM2012 isostatic gravity, EMAG2v3 upward-continued magnetic data, and mining-footprint counts. These layers were prepared during exploration but are not in the final DVHM feature set; the model uses fault distance, contact distance and class, Bouguer gravity, and the domain-varying terms.

2.3 Domain-level positive-observation structure and partial-pooling qualification

Cells with a compiled occurrence record in the modelling frame are concentrated exclusively in two of the six tectonic domains: 107 positive cells occur in NRB_3a (out of 1,127 total cells) and 31 occur in NRB_3b (out of 550 total cells). The remaining four domains contain zero recorded positive observations in this compilation: CRZ_4b ($n = 154$), NKB ($n = 24$), MMSB ($n = 16$), and SRB ($n = 1$).

Methodologically, this empirical structure has specific implications for the hierarchical Bayesian model:

- The formulation remains a fully partially pooled hierarchical model across all six domains, with domain-varying parameters linked through shared population hyperparameters μ_p and σ_p.
- In CRZ_4b, NKB, SRB, and MMSB, the likelihood contains no within-domain positive-class information. Their domain-specific effects are informed by the available negative observations together with the shared hierarchical population structure, and therefore rely more strongly on partial pooling than effects in domains containing both classes.
- Consequently, domain-specific coefficient interpretations for CRZ_4b, NKB, SRB, and MMSB must be treated with substantial caution, and standard binary classification metrics such as ROC-AUC are undefined within these single-class partitions.

2.4 Complete preprocessing chain

The overall modelling sequence is summarized in Figure 3. The prepared spatial frame contains 5 km grid cells with six Daly-domain categories (27 cells are unassigned). The four contiguous folds are assigned before model fitting. For each fold, continuous predictors are standardized using the training partition only, and the resulting transformation is applied unchanged to both training and held-out cells. Quadratic fault- and contact-distance terms are formed from the standardized values. Lithology categories are retained only when represented by both positive and background cells in the training partition; held-out data do not determine that support.

2.5 Distinction between validation folds and tectonic domains

This study uses two distinct spatial stratifications:

- **Along-belt validation folds (Folds 1–4):** Centroid coordinates are subjected to PCA; observations are projected onto the first principal component, sorted by that projection, and divided into four approximately equal contiguous groups of 468 cells. Each group is held out once to evaluate geographic transfer. This reduces local train-test overlap but does not guarantee independence across fold boundaries.
- **Daly tectonic domains (6 domains):** A geological and structural stratification based on Katangan basin evolution (Daly et al., 2025), used as the grouping variable for hierarchical partial pooling of model parameters.

The model is trained on three spatial folds and evaluated on the held-out fold; domain-level performance metrics are computed post hoc as descriptive strata. The spatial partition and the four held-out sectors are shown in Figure 2.

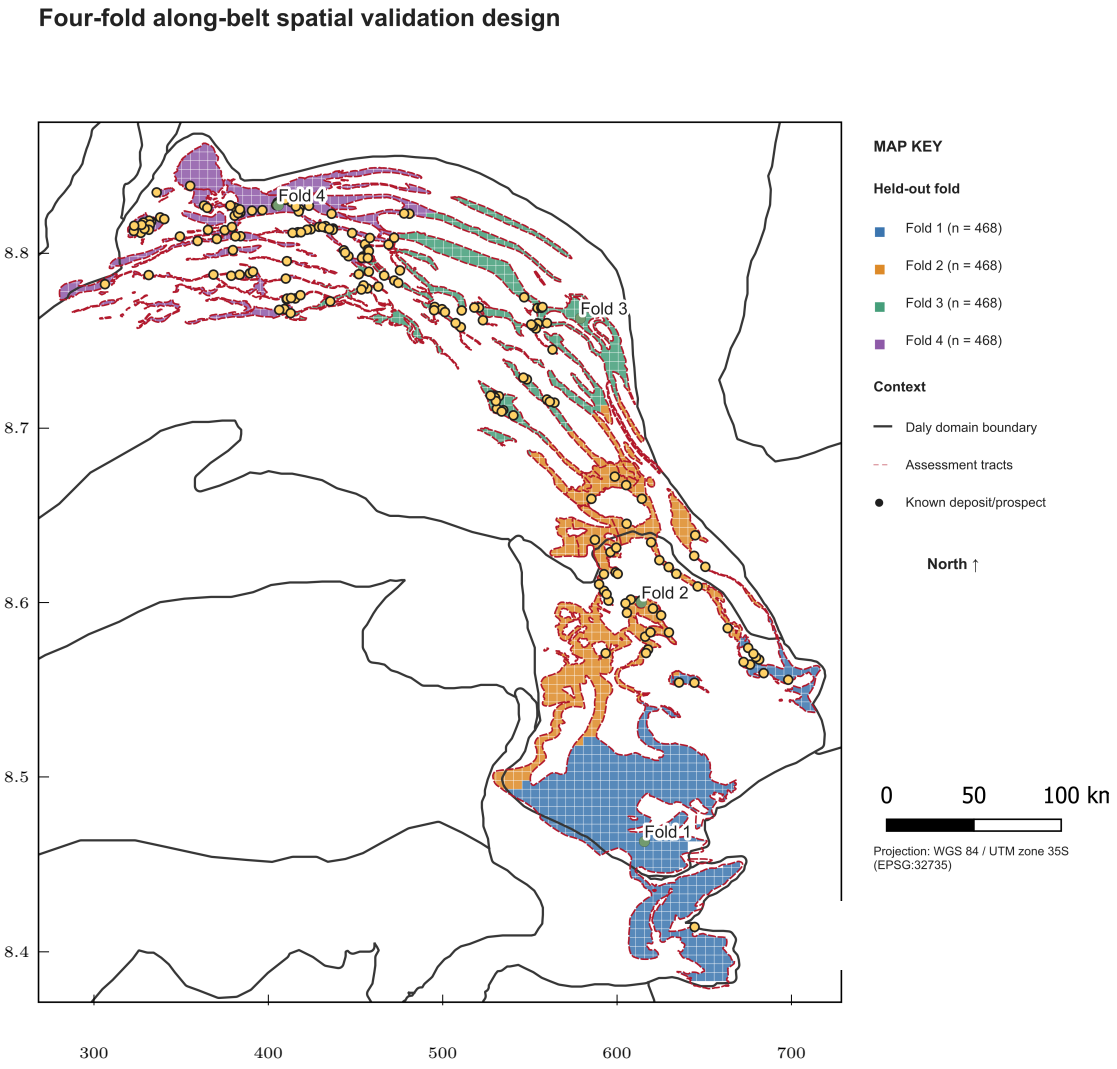


Figure 2: Four contiguous along-belt validation folds across the 5 km modelling grid, with 468 cells in each fold. Each fold is held out once while the other three are used for training. Tectonic-domain and provisional mineral-assessment tract outlines and compiled occurrence locations provide regional context; the fold assignment is distinct from the six Daly tectonic domains used for hierarchical grouping. Border ticks show UTM eastings in 10^3 m and northings in 10^6 m (EPSG:32735).

3 Bayesian hierarchical methodology and spatial validation

3.1 Domain-varying hierarchical model specification

For cell i in Daly domain $d(i) \in \{1, \ldots, 6\}$, the DVHM is a partially pooled hierarchical Bayesian logistic regression (Gelman and Hill, 2007):

$$Y_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \eta_i, \tag{1}$$

where the linear predictor η_i decomposes as:

$$\eta_i = \alpha_{d(i)} + \beta_{f,\text{lin},d(i)} z_{f,i} + \beta_{f,\text{sq},d(i)} z_{f,i}^2 + \beta_{l,\text{lin},d(i)} z_{l,i} + \beta_{l,\text{sq},d(i)} z_{l,i}^2 + \beta_g z_{g,i} + \mathbf{x}_{\text{rock},i}^{\mathsf{T}} \boldsymbol{\beta}_{\text{rock}}. \tag{2}$$

The model distinguishes:

- **Domain-varying parameters:** Domain intercepts α_d and the four distance coefficients (fault linear $\beta_{f,\text{lin},d}$, fault quadratic $\beta_{f,\text{sq},d}$, lithological contact linear $\beta_{l,\text{lin},d}$, and lithological contact quadratic $\beta_{l,\text{sq},d}$).
- **Global parameters:** The Bouguer gravity coefficient β_g and the surficial-lithology vector $\boldsymbol{\beta}_{\text{rock}}$ are shared globally across all domains.

Domain-varying parameters use a non-centred hierarchical parameterization to ensure efficient MCMC exploration:

$$\beta_{p,d} = \mu_p + \sigma_p \cdot \text{offset}_{p,d}, \qquad \text{for } p \in \{f_{\text{lin}}, f_{\text{sq}}, l_{\text{lin}}, l_{\text{sq}}\}, \tag{3}$$

and

$$\alpha_d = \mu_\alpha + \sigma_\alpha \cdot \text{offset}_{\alpha,d}. \tag{4}$$

The model uses the following priors:

$$\mu_p \sim \mathcal{N}(0, 1), \qquad \sigma_p \sim \text{HalfNormal}(1), \tag{5}$$

$$\text{offset}_{p,d} \sim \mathcal{N}(0, 1), \qquad \mu_\alpha \sim \mathcal{N}(\text{logit}(\hat{p}_{\text{train}}), 1)\,, \tag{6}$$

$$\sigma_\alpha \sim \text{HalfNormal}(1), \qquad \text{offset}_{\alpha,d} \sim \mathcal{N}(0, 1), \tag{7}$$

$$\beta_g \sim \mathcal{N}(0, 1), \qquad \boldsymbol{\beta}_{\text{rock}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{8}$$

where $\hat{p}_{\text{train}} = \sum y_{\text{train}} / N_{\text{train}}$ anchors the intercept prior to the empirical training base rate.

3.2 Continuous probability prediction and threshold-free evaluation

For each posterior draw $s \in \{1, \ldots, S\}$ and held-out cell i, the draw-specific parameters and the held-out cell's fold-preprocessed features determine its conditional prospectivity probability:

$$\eta_i^{(s)} = \alpha_{d(i)}^{(s)} + \ldots, \qquad p_i^{(s)} = \text{logit}^{-1}(\eta_i^{(s)}) = \frac{1}{1 + \exp(-\eta_i^{(s)})}. \tag{9}$$

The reported out-of-fold prediction is the posterior mean conditional probability:

$$\bar{p}_i = \frac{1}{S} \sum_{s=1}^{S} p_i^{(s)}. \tag{10}$$

No classification threshold is applied to generate predictions or compute discrimination metrics. Performance is evaluated across the continuous distribution of predicted probabilities using rank discrimination metrics that do not require a threshold (Lobo et al., 2008).

3.3 Uncertainty concepts

Four distinct uncertainty concepts apply to this analysis:

1. **Parameter posterior uncertainty:** Represented by MCMC posterior samples of coefficients and hyperparameters.
2. **Posterior uncertainty in the conditional probability:** For each held-out cell, the posterior distribution of p_i is induced by parameter uncertainty. Posterior means, medians, standard deviations, and 95% credible intervals quantify uncertainty in p_i, not whether a deposit will occur.
3. **Future outcome uncertainty:** The workflow produces conditional probability estimates, not simulated Bernoulli realizations $Y_i^{(s)} \sim \text{Bernoulli}(p_i^{(s)})$.
4. **Performance metric uncertainty:** Fold and pooled metric intervals use paired row-bootstrap resampling of fixed OOF scores; a separate multiscale spatial block bootstrap evaluates robustness of ΔAUC at specified block scales.

Throughout the results and discussion, fitted coefficients and predictor-component shares describe posterior statistical associations, not causal geological effects. This distinction is stated once here and applies to all subsequent reporting.

3.4 Decomposition of linear predictor variance

To describe how additive predictor components vary across held-out cells in each sector, the out-of-fold linear predictor is decomposed for each test cell i in fold k:

$$\eta_i = c_{\text{intercept},i} + c_{\text{fault},i} + c_{\text{lith},i} + c_{\text{grav},i} + c_{\text{rocks},i}, \tag{11}$$

where each component represents the posterior expectation across retained MCMC draws:

$$c_{\text{fault},i} = \mathbb{E}_{\text{post}}\left[\beta_{f,\text{lin},d(i)} z_{f,i} + \beta_{f,\text{sq},d(i)} z_{f,i}^2\right], \tag{12}$$

$$c_{\text{lith},i} = \mathbb{E}_{\text{post}}\left[\beta_{l,\text{lin},d(i)} z_{l,i} + \beta_{l,\text{sq},d(i)} z_{l,i}^2\right], \tag{13}$$

$$c_{\text{grav},i} = \mathbb{E}_{\text{post}}[\beta_g z_{g,i}], \tag{14}$$

$$c_{\text{rocks},i} = \mathbb{E}_{\text{post}}\left[\mathbf{x}_{\text{rock},i}^{\mathsf{T}} \boldsymbol{\beta}_{\text{rock}}\right]. \tag{15}$$

For each predictor component $m \in \{\text{fault}, \text{lith}, \text{grav}, \text{rocks}\}$, its dispersion across the N_{test} held-out cells of fold k is measured by its sample standard deviation:

$$\text{SD}_m = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i \in \mathcal{D}_{\text{test}}^{(k)}} (c_{m,i} - \bar{c}_m)^2}. \tag{16}$$

The reported component share is the SD-normalized share of additive-component dispersion:

$$\text{Share}_m = \frac{\text{SD}_m}{\text{SD}_{\text{fault}} + \text{SD}_{\text{lith}} + \text{SD}_{\text{grav}} + \text{SD}_{\text{rocks}}} \times 100\%. \tag{17}$$

These shares describe relative dispersion of additive components across held-out cells and are not comparable to proportions of variance explained.

3.5 Distance-representation sensitivity design and robustness criteria

To test whether distance-response inference depends on mathematical representation, four models form a 2×2 factorial grid:

Table 2: Distance-representation sensitivity design (2×2 grid).

	Quadratic	Linear
Raw distance (z)	Model A (DVHM)	Model C
Log distance ($\log(1 + x_{\text{km}})$)	Model B	Model D

Model A is the primary DVHM specification with raw quadratic distance terms; Models B–D are the corresponding sensitivity variants. Models A and C use standardized raw distance; Models B and D use standardized $\log(1 + x_{\text{km}})$.

As a non-hierarchical comparator, a compact global logistic baseline (CGLB) is fitted separately within each training fold under the same four-fold design. The CGLB contains standardized Bouguer gravity and lithological-contact distance with its quadratic term; it includes neither fault distance nor domain-varying parameters.

Across the four spatial folds, six Daly domains, and two distance predictors (fault and lithological contact), there are $4 \times 6 \times 2 = 48$ evaluable comparison cases. In each case, Model A (raw-quadratic) is compared against Model B (log-quadratic) using three pre-registered criteria:

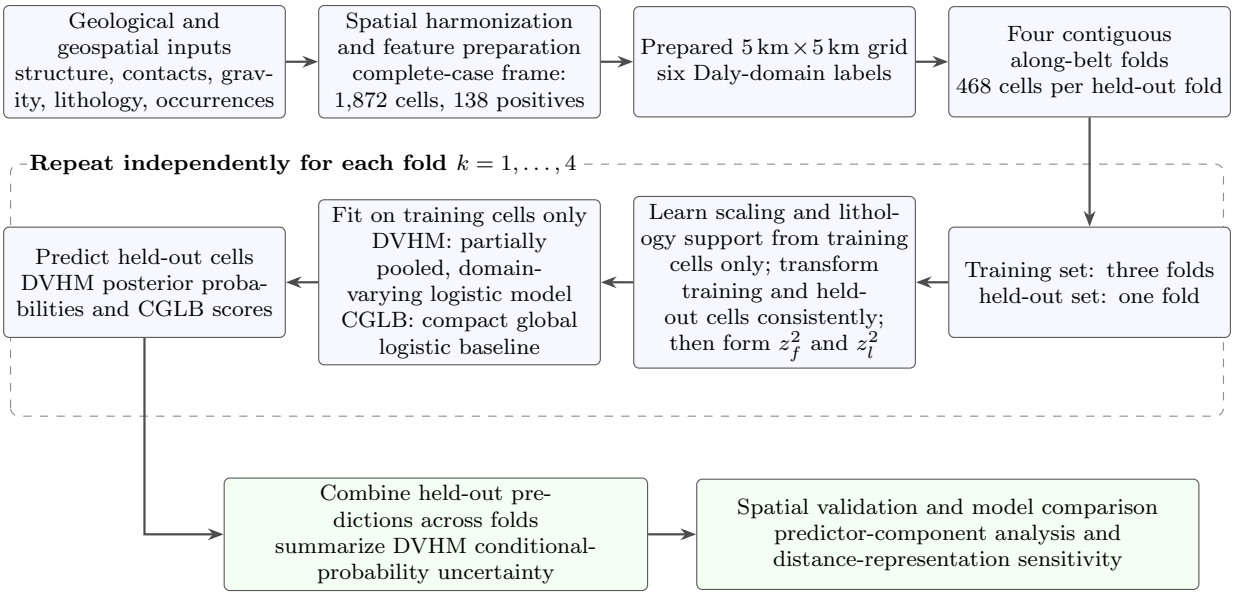


Figure 3: Modelling workflow. The six Daly domains define the hierarchical grouping, whereas the four contiguous folds define validation. Within each fold, scaling and lithology support are learned from training cells only; quadratic distance terms are constructed after standardization. Both models are fitted on training cells, and predictions are made only for the held-out fold before out-of-fold results are combined for evaluation.

1. **Criterion 1 (Curvature probability agreement):** The absolute difference in posterior probability of positive quadratic curvature between representations must be less than 0.15:

$$|P(\beta_{\text{sq}} > 0 \mid \text{Model A}) - P(\beta_{\text{sq}} > 0 \mid \text{Model B})| < 0.15. \tag{18}$$

2. **Criterion 2 (D^* median agreement):** The relative difference between posterior median stationary points D_A^* and D_B^* (in physical kilometers) must not exceed 25%:

$$\frac{|D_A^* - D_B^*|}{\max(|D_A^*|, |D_B^*|)} \leq 0.25. \tag{19}$$

3. **Criterion 3 (Data support concentration):** Both median stationary points must fall within their respective training data empirical support ranges:

$$\text{support_min}_A \leq D_A^* \leq \text{support_max}_A \quad \text{and} \quad \text{support_min}_B \leq D_B^* \leq \text{support_max}_B. \tag{20}$$

A relationship is classified as representationally robust if and only if it satisfies all three criteria simultaneously.

4 Results

4.1 Spatial predictive transferability and prospectivity mapping

The DVHM out-of-fold predictions produce the regional prospectivity map in Figure 4. Across the four folds, DVHM ROC-AUC values are 0.817, 0.688, 0.676, and 0.561, and PR-AUC values

are 0.048, 0.103, 0.133, and 0.145. Pooled across all 1,872 cells, DVHM ROC-AUC is 0.689 (95% bootstrap CI: 0.647–0.727), compared with 0.525 (0.470–0.581) for CGLB, a gain of +0.164 (+0.084 to +0.236).

Pooled performance masks geographic divergence. In Fold 3, the DVHM exceeds the CGLB by $\Delta\text{AUC} = +0.091$ (95% paired row-bootstrap CI $[+0.024, +0.159]$); in Fold 2, the DVHM trails the CGLB by $\Delta\text{AUC} = -0.136$ (95% CI $[-0.250, -0.020]$). The corresponding intervals for Folds 1 and 4 span zero.

In secondary domain stratification, NRB_3a DVHM ROC-AUC is 0.662 versus 0.536 for CGLB ($\Delta = +0.126$). In NRB_3b, DVHM ROC-AUC is 0.584 versus 0.867 for CGLB ($\Delta = -0.283$). ROC-AUC is undefined in CRZ_4b, NKB, SRB, and MMSB because these strata contain no positive observations.

4.2 Multivariate posterior structure

How do predictor associations differ across domains? In NRB_3a, lithological-contact distance has negative linear coefficient medians (−0.849 to −0.451) and positive quadratic coefficient medians (+0.293 to +0.334), with posterior probabilities of positive curvature $P(\beta_{l,\text{sq}} > 0) \geq 0.993$ across all four folds. In NRB_3b, posterior support for positive curvature is fold-dependent and weaker than in NRB_3a.

Fault-distance effects vary more dramatically. In NRB_3a, linear medians range from −2.681 to −0.703, while quadratic medians are negative in Fold 2 and positive in Folds 1, 3, and 4. In NRB_3b, fault linear medians shift to positive values (+0.181 to +1.548), demonstrating that the directional association with fault distance is not spatially uniform across the belt.

4.3 Spatial heterogeneity in predictor contributions

Which predictor components drive the most variation in the out-of-fold linear predictor, and does their relative importance shift across sectors? The SD-normalized shares (Table 3 and Figure 5) reveal a clear along-belt gradient:

Table 3: SD-normalized shares of decomposed out-of-fold linear-predictor components.

Fold	Fault distance	Lithology distance	Bouguer gravity	Surficial lithology
1	86.5%	11.1%	2.0%	0.4%
2	50.3%	23.9%	5.6%	20.3%
3	33.1%	28.2%	15.3%	23.4%
4	34.9%	17.7%	31.1%	16.3%

Fault distance dominates in Fold 1 (86.5%) and diminishes southward (50.3%, 33.1%, 34.9%), while Bouguer gravity rises from 2.0% in Fold 1 to 31.1% in Fold 4. Surficial lithology and contact distance have their largest shares in Folds 2 and 3.

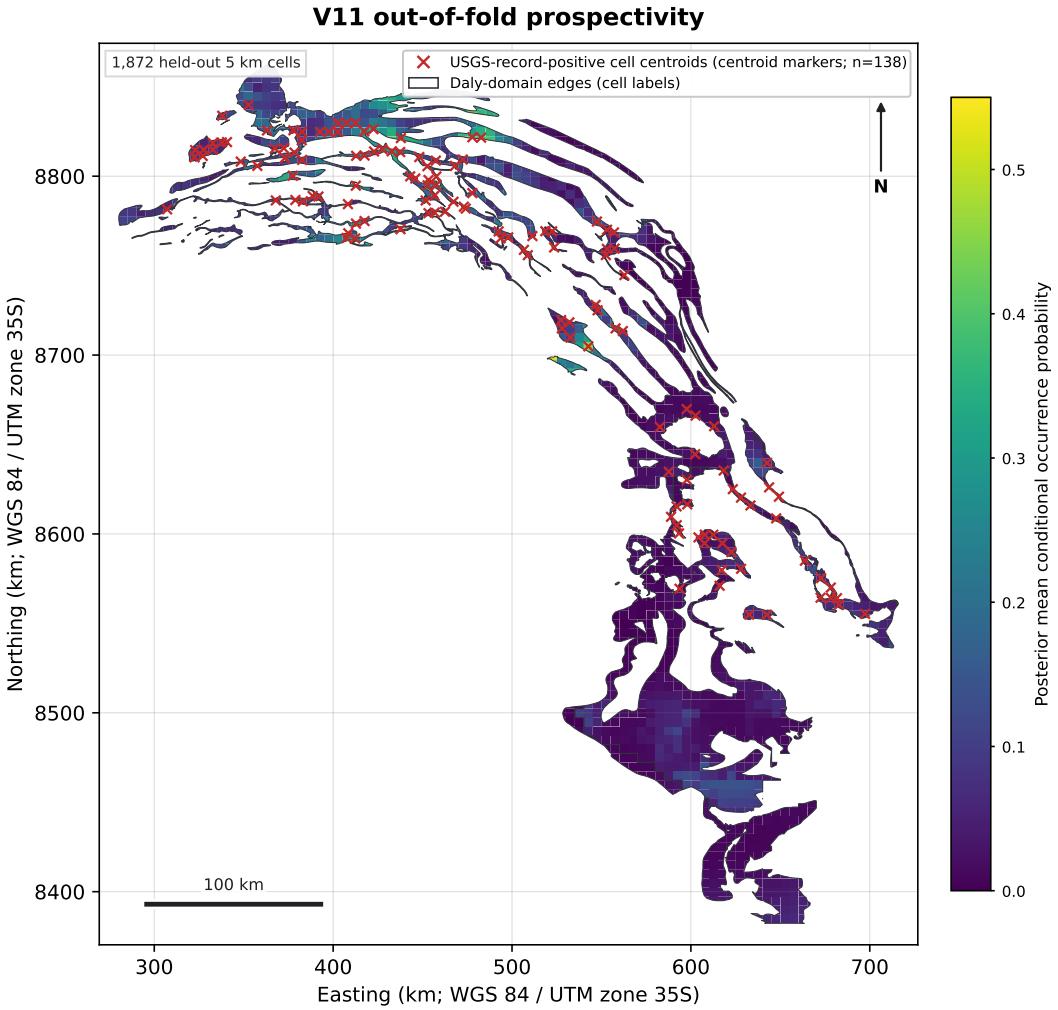


Figure 4: Out-of-fold map of posterior mean conditional probability of a compiled occurrence-present cell under the DVHM. The modelling-cell polygons are shaded by $\bar{p}_i$; red crosses mark centroids of the 138 cells labeled positive in the complete-case model, not the exact locations of source points within those cells. Fine outlines show Daly-domain edges associated with the stored cell-level domain labels. Observed cell labels are distinguished from model predictions. The color scale reports conditional probability of the compiled cell label, not probability of discovery. Coordinates are WGS 84 / UTM zone 35S (EPSG:32735); the x and y scale bar denotes 100 km. Cell-level 95% posterior credible-interval widths are provided separately in the accompanying uncertainty map and GeoPackage.

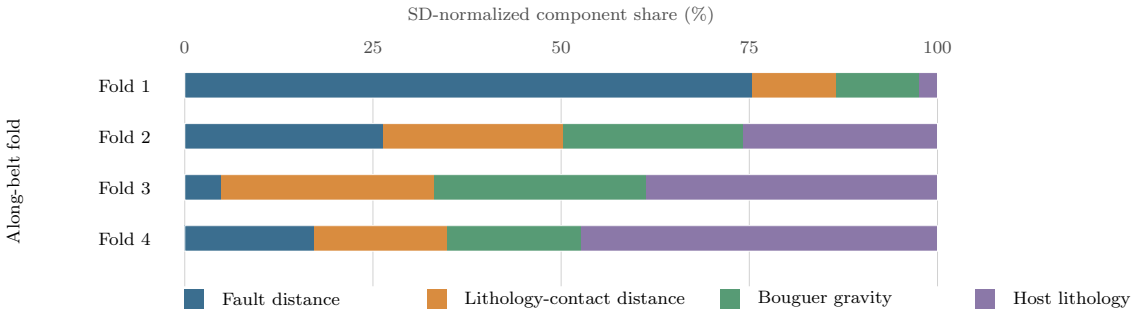
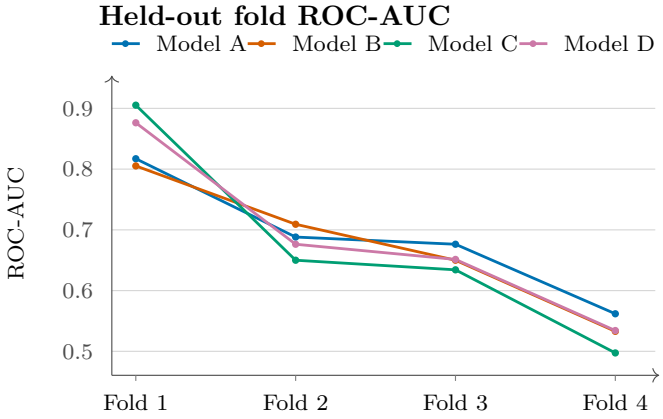


Figure 5: SD-normalized shares of dispersion in additive components of the out-of-fold linear predictor across four spatial folds. The stacked segments show relative component shares; these are SD-normalized shares, not proportions of outcome variance explained.



Mean across folds

Model	ROC-AUC	PR-AUC
Model A	0.686	0.107
Model B	0.673	0.123
Model C	0.671	0.130
Model D	0.683	0.145

Robustness criteria (48 cases): curvature-sign agreement, 13/48; D^* median agreement, 0/48; support concentration, 15/48; all three criteria, 0/48.

Figure 6: Sensitivity of held-out discrimination to the four distance representations. The left panel shows fold-level ROC-AUC; the right panel reports mean ROC-AUC and PR-AUC across folds.

4.4 Distance-representation sensitivity and non-identifiability of D^*

Figure 6 summarizes fold-level discrimination and the mean ROC-AUC and PR-AUC for the four distance representations (Table 2). Mean ROC-AUC values are 0.686 (Model A), 0.673 (B), 0.671 (C), and 0.683 (D). Mean PR-AUC values are 0.107, 0.123, 0.130, and 0.145 respectively, with the log-linear specification (Model D) achieving the highest mean PR-AUC.

While proximity associations persist, polynomial curvature and the derived stationary point $D^* = -\beta_{\text{lin}}/(2\beta_{\text{sq}})$ are representation-dependent. Across the 48 evaluation cases, curvature-sign agreement held in 13, stationary-point agreement in 0, and posterior support concentration in 15. All three criteria were satisfied simultaneously in **0 of 48 cases**.

At the population level, posterior curvature distributions overlap zero: $P(\mu_{f,\text{sq}} > 0) = 0.384$ and $P(\mu_{l,\text{sq}} > 0) = 0.743$. The 95% credible intervals for population D^* span $[-279.36, 366.38]$ km for fault distance and $[-161.69, 152.34]$ km for lithological contact distance. These intervals include negative distances and values far exceeding the basin width, demonstrating that population-level D^* is not identifiable.

5 Discussion

5.1 Predictor associations and conditional inference

The multivariate results establish that geological and geophysical predictors contain joint predictive signal in the Copperbelt modelling frame. Fault distance, lithological-contact distance, Bouguer gravity, and surficial lithology contribute jointly to the fitted out-of-fold predictor. The hierarchical structure permits domain-varying intercepts and distance effects while sharing information across Daly domains through partial pooling; individual coefficients therefore reflect associations conditional on the other predictors.

5.2 Spatial non-stationarity and geographic transferability

The OOF decomposition shows that predictor-component shares shift systematically along the belt: fault distance dominates in the north, while Bouguer gravity becomes co-dominant in the south. The regional geological literature documents corresponding variation in basin structure and stratigraphic setting (Cailteux et al., 2005; Selley et al., 2005; Hitzman et al., 2012; Ndonfack et al., 2025), providing context for interpreting these patterns.

Spatially separated prediction demonstrates that pooled belt-wide skill can conceal regional performance divergence. The DVHM outperforms the CGLB in Fold 3 but underperforms it in Fold 2, reinforcing the necessity of spatially separated validation strategies (Roberts et al., 2017; Valavi et al., 2019). Daly's tectonic domain classification (Daly et al., 2025) provides a principled geological stratification for hierarchical partial pooling, but the analysis estimates empirical associations within the compiled data and does not constitute a formal test of Daly's tectonic hypothesis.

5.3 Representation sensitivity and model parsimony

Proximity associations persist across mathematical transformations, but polynomial curvature and the derived stationary point D^* are sensitive to functional form. The zero-pass rate across the 48 robustness cases demonstrates that D^* is a model-dependent algebraic property of quadratic curves rather than an invariant physical quantity, and prospectivity mapping should avoid translating quadratic extrema into drilling targets.

The comparable discrimination achieved by linear and log-linear specifications (Models C and D) relative to quadratic models indicates that quadratic distance terms did not improve mean discrimination. Linear and logarithmic representations are simpler specifications; however, the results do not establish a universally preferred distance form.

5.4 Limitations

Several methodological limitations apply:

1. **Sample size and domain imbalance:** Only 138 cells carry a compiled occurrence label, with 107 concentrated in NRB_3a and 31 in NRB_3b. Four domains contain zero positive labels, precluding within-domain discrimination evaluation.

2. **Residual spatial autocorrelation:** Contiguous spatial blocking mitigates optimistic cross-validation bias, but spatial autocorrelation may persist across block boundaries (Roberts et al., 2017).

3. **Input layer uncertainty:** Regional geological maps, structural compilations, and gravity grids contain measurement, interpolation, and cartographic errors that propagate into distance calculations.

4. **Surface proxies vs. 3D geometry:** Near-surface distance features do not capture three-dimensional fault dip, subsurface stratigraphy, or depth to basement.

5. **Discrete blocking approximation:** Four discrete spatial folds approximate a continuous along-belt metallogenic gradient.

6 Conclusions

The DVHM provides uncertainty-aware out-of-fold conditional-probability estimates for 1,872 modelling cells across the Central African Copperbelt. Posterior uncertainty is propagated to each held-out cell, yielding posterior means, standard deviations, and 95% credible intervals for conditional occurrence probabilities.

The OOF decomposition reveals geographically varying predictor-component shares: fault-distance dominance in the northern sectors gives way to co-dominant Bouguer gravity in the south. Fold-level and domain-level discrimination vary substantially, demonstrating that a pooled belt-wide metric can conceal regional performance divergence.

Distance associations recur across the evaluated representations, but quadratic curvature and the derived D^* diagnostics are representation-dependent, and population-level D^* is not identifiable. Spatially separated validation evaluates transfer among the specified folds but does not guarantee transfer to all frontier areas.

7 Reproducibility, data, and code availability

The analyses use PyMC (Salvatier et al., 2016), ArviZ, and scikit-learn. The DVHM was estimated using two chains, 2,500 tuning steps, and 1,500 retained draws per chain, with target acceptance 0.99 and random seed 42. Code, available prepared data, and derived results are provided at `https://github.com/vmtasi/copperbelt-mineral-prospectivity`. The USGS occurrence compilation, the BGI gravity product, and the Africa surficial lithology raster are cited above but are not redistributed with the repository. The modelling frame is reproducible as the intersection of the 5 km grid with the five USGS permissive tracts. The frozen fault, gravity, and occurrence values are numerically reproduced from archived layers; the contact class and distance are reproduced from the archived many-to-many table but not from a recorded nearest-contact rule. The exact QGIS operation sequence, the provenance and accuracy of the fault-map control points, the contact-table row order, and the domain polygon digitization and gap/overlap resolution were not recorded in the available processing history. These limits should be considered when interpreting the spatial predictors; they do not alter the frozen model inputs or predictions reported here.

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